Ecuaciones de capacidad de soporte

Hansen (1970).* See Table 4-5 for shape, depth, and other factors.

General:† $q_{ult} = cN_c s_c d_c i_c g_c b_c + \bar{q} N_q s_q d_q i_q g_q b_q + 0.5\gamma B' N_\gamma s_\gamma d_\gamma i_\gamma g_\gamma b_\gamma$

when $\phi = 0$

use $q_{ult} = 5.14 s_u (1 + s'_c + d'_c - i'_c - b'_c - g'_c) + \bar{q}$

N_q = same as Meyerhof above

N_c = same as Meyerhof above

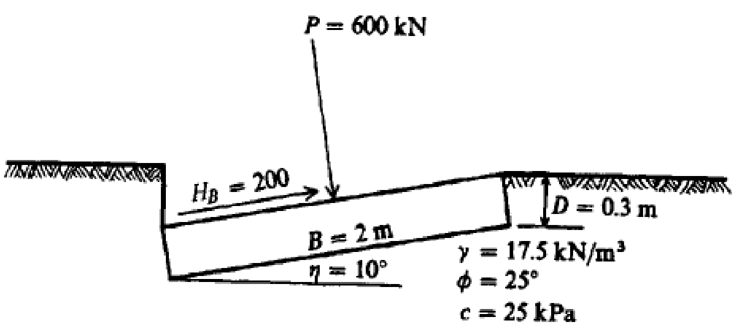
$N_\gamma = 1.5(N_q - 1)\tan\phi$

Vesić (1973, 1975).* See Table 4-5 for shape, depth, and other factors.
Use Hansen's equations above.

N_q = same as Meyerhof above

N_c = same as Meyerhof above

$N_\gamma = 2(N_q + 1)\tan\phi$



Ground factors (base on slope)

Hansen

$$g'_c = \frac{\beta^\circ}{147^\circ}$$

$$g_c = 1.0 - \frac{\beta^\circ}{147^\circ}$$

$$g_q = g_\gamma = (1 - 0.5\tan\beta)^5$$

Base factors (tilted base)

$$b'_c = \frac{\eta^\circ}{147^\circ} \quad (\phi = 0)$$

$$b_c = 1 - \frac{\eta^\circ}{147^\circ} \quad (\phi > 0)$$

$$b_q = \exp(-2\eta\tan\phi)$$

$$b_\gamma = \exp(-2.7\eta\tan\phi)$$

η in radians

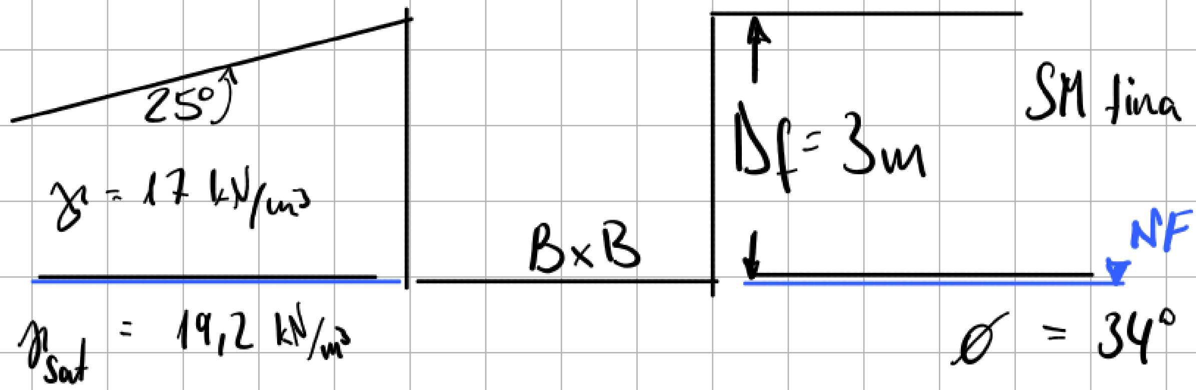
1)

Perfil Estratigráfico

[m]

[m]				
0 – 1	1m	???	? ?? ?	?
1 – 2	1m	Escombros	$\gamma = 14 \, kN/m^3$	$\phi = 32^\circ$
2 – 3	1m	Maicillo	$\gamma = 18 \, kN/m^3$	$\phi = 34^\circ$
3 – 4	1m	Arena limosa (SM)	$\gamma = 19 \, kN/m^3$	$\phi = 38^\circ$
4 – 6	2m	Arena fina (SP)	$\gamma = 19 \, kN/m^3$	$\phi = 41^\circ$
6 – 7	1m	Arena media (SP)	$\gamma = 19{,}5 \, kN/m^3$	$\phi = 46^\circ$
7 – 8	1m	Arena fina (SP)	$\gamma = 19 \, kN/m^3$	($\phi = 41^\circ$ Supuesto)
8 – 11	3m	Arcilla baja plasticidad (CL)	$\gamma = 14{,}5 \, kN/m^3$	$\phi = 34^\circ$
11 –		Arena fina mal graduada (SP)	$\gamma = 18 \, kN/m^3$	$\phi = 41^\circ$

Problema 2



$$q_{adm} = 10 kPa \times Piso$$

a) q_{ult} Hansen:

- $N_c = 42{,}16$; $N_q = 29{,}44$; $N_\gamma = 28{,}77$
- $S_c = 1{,}69$ $\quad S_q = 1{,}56$ $\quad S_\gamma = 0{,}6$

Si $K > 1 \Rightarrow B < 3$ (Zapatas) ; $k = Tan^{-1}(3/B)$

- $d_c = 1{,}52$ $\quad d_q = 1{,}34$ $\quad d_\gamma = 1$
- No hay carga inclinada → No $i_{c,q,\gamma}$
- $g_c = 0{,}83$ $\quad g_q = 0{,}76$ $\quad g_\gamma = 0{,}76$
- Suelo bajo fundación no inclinado → No $g_{c,q,\gamma}$

$$q_{ult} = \bar{q} \cdot N_q' \cdot S_q \cdot d_q \cdot g_q + 0{,}5 \cdot \gamma \cdot B \cdot N_\gamma' \cdot S_\gamma \cdot d_\gamma \cdot g_\gamma$$

$$N_\gamma' = \frac{N_\gamma}{2} + \frac{N_\gamma}{2}\left[R + \frac{b}{2B}(1-R)\right]$$

$$\boxed{\begin{array}{l} \text{con } B = 2m \\ b/B = 0 \\ D/B = 1{,}5 \end{array}}$$

$R = k_{min}/k_{max}$ $\quad k_{min} = 2{,}63$ y $k_{max} = 1023$

$N_\gamma' = 7{,}85$ $\quad N_q' = 36{,}72$ $\quad N_c' = 41{,}56$

$q_{ult} = 3189\ kPa$; $q_{adm} \approx 800\ kPa$

En una zapata cuadrada de 2 m su carga max. Seria de $q_{adm} \times \text{Area}_{zapata} = 800 kPa \times 4 m^2 = \underline{3200\ kN}$

Supuesto : * El edificio tiene $100 m^2$
* La zapata está en el centro

$10 kPa \times$ Piso $= \underline{1000 kN} \times$ Piso

Conclusión : La zapata aguantaria 3 pisos

Hansen:

$$q_{ult} = c N'_c S_c i_c + \bar{q} N'_q S_q i_q + \gamma \frac{B'}{2} N'_\gamma S_\gamma i_\gamma$$

- Se considera que B sera de $2{,}65$ m
- De ensayo tenemos $\phi = 40°$ $\to b = 0$

$N_\gamma = 79{,}54 \rightsquigarrow N'_\gamma = \frac{N_\gamma}{2} = 39{,}77$

$S_c = 1{,}85$; $S_q = 1{,}64$; $S_\gamma = 0{,}6$

$i_c = 0{,}83$ $i_q = 0{,}835$ $i_\gamma = 0{,}77$

- de tabla 4,7 con $\phi = 40°$ $\frac{D}{B} = 0{,}75$ $\frac{b}{B} = 0$

$N'_c = 62{,}18$ $N'_q = 25{,}09$

$q_{ult} = 2580 \text{ kPa} \rightsquigarrow q_{adm} = \frac{q_{ult}}{FS} = 860 \text{ kPa}$

Conclusión: Ya que $B = 2{,}65$

$V_{max} = q_{adm} \cdot A_F = 6040 \text{ kN} > V_{diseño} = 2500 \text{ kN}$

Por lo que una zapata de 2,65 m es válida